

AI Video Forensic Lab (Process 5.0)

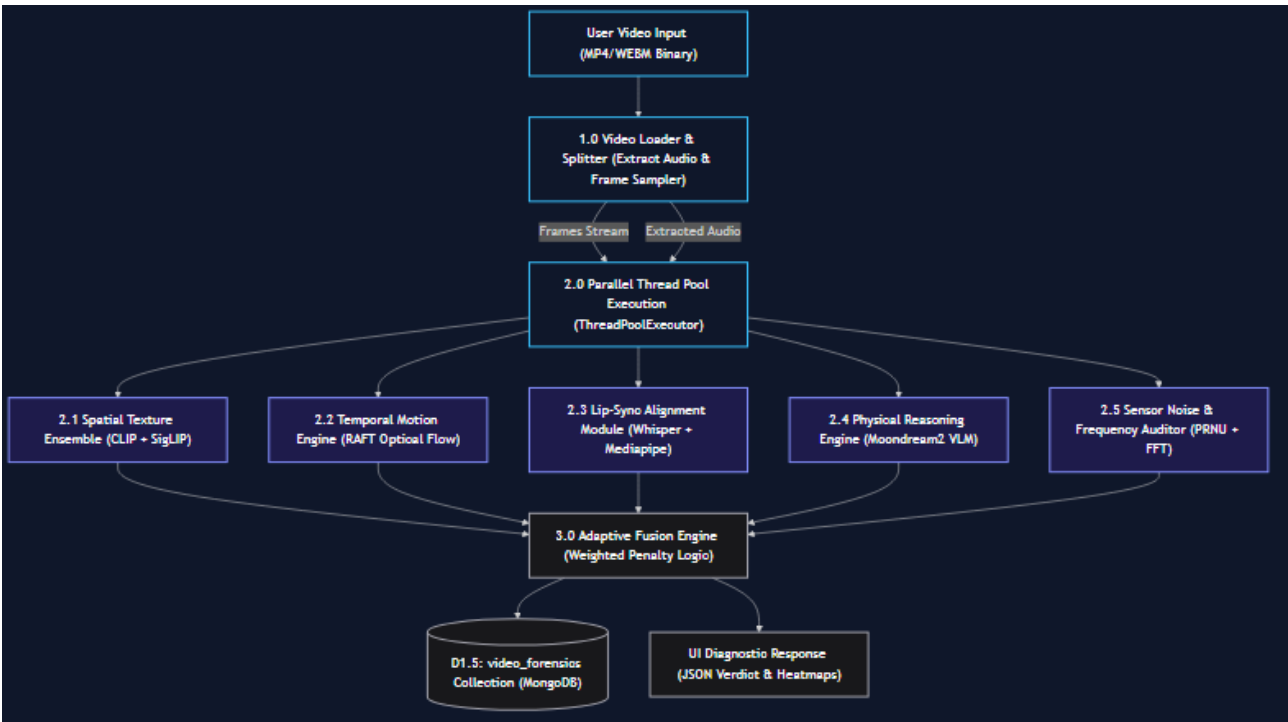
System Architecture, Phased Pipeline, Forensic Model Ensemble, and Multi-modal Fusion

This document explains the technical implementation, model pipeline, and processing workflow of the **AI Video Forensic Lab (Process 5.0)** in the FakeShield system.

Video Lab Processing Architecture Diagram

The flowchart below demonstrates the sequential and parallel data flow from the moment a user uploads a video file down to database logging.

Figure 1: Video Lab Forensic Processing Workflow (v11.0)



Core Stages of the Video Forensic Pipeline

1. Video Loading & Preprocessing (`video_router.py`)

- **Frame Sampling:** To guarantee sub-5s processing on standard CPU workloads, the video router limits analysis to **8-16 keyframes** distributed evenly across the video length.
- **Audio Demuxing:** Uses `ffmpeg` or local sub-processes to isolate the audio track as a WAV file, which is mapped directly to the Lip-Sync module.

■■ Multi-Model Analyzer Specifications

■ Deep Learning Analyzers (Spatial, Temporal, & Reasoning)

A. Spatial Texture Ensemble (`video_clip.py`)

- **Models:** `openai/clip-vit-base-patch32` (weight: 40%) & `google/siglip-base-patch16-224` (weight: 60%).
- **Method:** Performs zero-shot classification at the frame level. Contrastive text prompt matrices are mapped to the images:
- **Real Prompts:** "a frame from a real video recorded by a camera", "natural video footage".
- **AI Prompts:** "a frame from an AI-generated synthetic video", "synthetic content from Sora/Runway".
- **Aggregator:** Rather than using a mean average (which dilutes localized frame edits like face swaps), it takes the **75th percentile** score.

B. Temporal Flow Engine (`video_tempoRAFT.py`)

- **Model:** `torchvision.models.optical_flow.raft_small` (pretrained).
- **Method:** Computes dense optical flow fields between consecutive frame pairs.
- **Heuristics:**
 1. **PAVR (Peak-to-Average Velocity Ratio):** Detects unnatural temporal morphing spikes.
 2. **Flow Entropy:** Measures spatial randomness of pixel movements.
 3. **Temporal Coherence Residual:** Computes the mean difference between consecutive flow vectors.
- **Artifact Output:** Converts raw optical flow tensors into BGR/RGB **HSV optical flow heatmaps** returned to the React frontend as base64 images.

C. Lip-Sync Alignment Module (`video_audio.py`)

- **Speech Detection:** Uses OpenAI's **Whisper ("base" model)** with greedy search parameters (`beam_size=1`, `best_of=1`) to extract precise word-level speaking timestamps from the audio track.
- **Lip Motion Tracking:** Uses **Mediapipe Tasks FaceLandmarker** (`face_landmarker.task`) to track lip openness over time by measuring the distance between upper lip indices [**61, 185, 40, ...**] and lower lip indices [**146, 91, 181, ...**] normalized by facial height (landmarks 10 to 152).
- **Correlation Mismatch:** Checks if lip movement aligns with the speech activity timeline. A high viseme-to-phoneme mismatch rate indicates dubbing or deepfake face synthesis.

D. Physical Reasoning Engine (`video_reasoning.py`)

- **Model:** **Moondream2** (`vikhyatk/moondream2` revision **2024-08-26**).
- **Method:** Evaluates geometric and physical inconsistencies in target frames (shadow errors, floating pixels, merging features).
- **Consolidated Forensic Prompt:**

"Analyze this frame for AI anomalies (warping, shadows, blurring). Keep response strictly under 15 words. End with exactly 'CONSISTENT' or 'INCONSISTENT'."

- **Decision Rule:** Scans the VLM output text for suspicious terms (**warp, ghost, blur, inconsistent**) to calculate the score.

■ Digital Signal Processing (DSP) Analyzers

E. Sensor Noise & Frequency Auditor (video_forensics_v2.py)

- **PRNU Consistency Check:** Photo Response Non-Uniformity (PRNU) acts as a camera sensor fingerprint. The module extracts frame noise residuals using a Gaussian high-pass filter:

Noise = Frame_gray - GaussianFilter(Frame_gray, sigma=2.0)

It calculates the cosine correlation coefficient between consecutive frame residuals. Real video shows high cross-correlation due to the shared physical camera sensor. AI-generated video (where frames are generated independently) displays zero correlation.

- **2D FFT Spectral Decay:** Applies a 2D Fast Fourier Transform (`np.fft.fft2`) on frames, mapping radial average power profiles. GAN/Diffusion generators leave periodic high-frequency checkerboard noise grid signatures, warping the expected $1/f^2$ spectral decay law.

■ The Adaptive Fusion Engine (video_fusion.py)

Unlike static classifiers, the Video Fusion Engine dynamically changes model weights based on the video's vertical resolution height (h):

1. Dynamic Weight Matrix

Resolution	Spatial (CLIP/SigLIP)	Temporal (RAFT)	Audio (Lip-Sync)	Forensic (PRNU/FFT)	Reasoning (VLM)
Low (<480p)	20%	20%	30%	5%	25%
Standard (480p - 720p)	25%	25%	20%	15%	15%
High (lge 1080p)	25%	30%	15%	20%	10%

Rationale: In low-resolution files, fine sensor noise details (PRNU/FFT) are washed out by compression, so the engine relies heavily on audio lip-sync and VLM reasoning. In high-resolution files, high-frequency spatial and temporal vectors are highly reliable.

2. Consistency Penalty

If the spatial score (static texture) and temporal score (flow consistency) disagree significantly ($|\text{Score}_{\text{spatial}} - \text{Score}_{\text{temporal}}| > 0.40$), the engine applies a **+0.10 penalty** to the final probability, reflecting cross-modal morphing instabilities.

3. Verdict Mappings

- $\geq 75\%$: **DEEPFAKE (CRITICAL Threat)**
- $55\% - 74\%$: **LIKELY FAKE (HIGH Threat)**
- $35\% - 54\%$: **UNCERTAIN (MEDIUM Threat)**
- $< 35\%$: **LIKELY REAL (LOW Threat)**